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MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION,
DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT
(PNS PROJECTS)
02/25

PART 1 GENERAL **AMENDMENT 008, 03-10-2026**

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP/SAFE Z9.2 (2018) Fundamentals Governing the Design
and Operation of Local Exhaust Ventilation
Systems

ASTM INTERNATIONAL (ASTM)

ASTM E1613 (2012) Standard Test Method for
Determination of Lead by Inductively
Coupled Plasma Atomic Emission
Spectrometry (ICP-AES), Flame Atomic
Absorption Spectrometry (FAAS), or
Graphite Furnace Atomic Absorption
Spectrometry (GFAAS) Techniques

ASTM E1644 (2017) Standard Practice for Hot Plate
Digestion of Dust Wipe Samples for the
Determination of Lead

ASTM E1726 (2001; R 2009) Preparation of Soil Samples
by Hotplate Digestion for Subsequent Lead
Analysis

ASTM E1727 (2016) Standard Practice for Field
Collection of Soil Samples for Subsequent
Lead Determination

ASTM E1728 (2016) Collection of Settled Dust Samples
Using Wipe Sampling Methods for Subsequent
Lead Determination

ASTM E1792 (2003; R 2016) Standard Specification for
Wipe Sampling Materials for Lead in
Surface Dust

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2019) Standard Methods of Fire Tests for
Flame Propagation of Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational
Health (SOH) Requirements

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780 (1995; Errata Aug 1996; Rev Ch. 7 - 1997)
Guidelines for the Evaluation and Control
of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1126	Chromium (VI)
29 CFR 1926.1127	Cadmium
29 CFR 1926.21	Safety Training and Education
29 CFR 1926.33	Access to Employee Exposure and Medical Records
29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1926.59	Hazard Communication
29 CFR 1926.62	Lead
29 CFR 1926.65	Hazardous Waste Operations and Emergency Response
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 745	Lead-Based Paint Poisoning Prevention in Certain Residential Structures
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements

49 CFR 178

Specifications for Packagings

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23

(2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586

(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

STATE OF MAINE, DEPARTMENT OF ENVIRONMENTAL PROTECTION (DEP)

ME L&C PROGRAM

State of Maine Licensing and Certification Program

STATE OF MAINE STATUTES AND REGULATIONS

Statutes and Regulations

Title 38, Chapter 12-B; Lead Abatement Emergency Provision §1296; Title 38, Chapter 12-B: Lead Abatement §1291-§1297; Title 22, Chapter 252: Lead Poisoning Control Act §1314-§1327; Title 14, Chapter 710: Environmental Lead Hazards Disclosure §6030-B; Rules; Chapter 424- Lead Management Regulations

PORTSMOUTH NAVAL SHIPYARD (PNSY)

5090.6F

(4-21) Solid Waste Operations

5090.30

(4-05) Hazardous Waste Generator Standards

5090.8

Environmental Sampling Manual

1.2 DEFINITIONS

1.2.1 Abatement

Measures defined in 40 CFR 745, Section 223, designed to permanently eliminate lead-based paint hazards.

1.2.2 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.3 Area Sampling

Sampling of lead, cadmium, and chromium (VI) concentrations within the lead, cadmium, and chromium (VI) control area and inside the physical boundaries which is representative of the airborne lead, cadmium, and

chromium (VI) concentrations but is not collected in the breathing zone of personnel (approximately 5 to 6 feet above the floor).

1.2.4 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.5 Category A LCC Work

Work operations involving Lead, Chromium, or Chromium (VI) containing material where the concentration of airborne lead particles is less than the AL.

1.2.6 Category B LCC Work

Work operation involving Lead, Chromium, or Chromium (VI) containing material where the concentration of airborne lead particles is equal to or greater than the AL but less than the PEL.

1.2.7 Category C LCC Work

Work operation involving Lead, Chromium, or Chromium (VI) containing material where the concentration of airborne lead particles is likely to exceed the PEL.

1.2.8 Certified Industrial Hygienist (CIH)

As used in this section refers to a person retained by the Contractor who is certified as an industrial hygienist and who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current federal, State, and local regulations. CIH must be certified for comprehensive practice by the American Board of Industrial Hygiene. The Certified Industrial Hygienist must be independent of the Contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.2.9 Child-Occupied Facility

Real property which is a building or portion of a building constructed prior to 1978 visited regularly by the same child, six-years of age or under, on at least two different days within any week (Sunday through Saturday period), provided that each day's visit lasts at least 3 hours, and the combined annual visits last at least 60 hours. Child-occupied facilities include but are not limited to, day-care centers, preschools and kindergarten classrooms.

1.2.10 Chromium (VI) Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1126. If an employee is exposed for more than 8 hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

1.2.11 Competent Person (CP)

As used in this section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current federal, State, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium (VI) hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene or a Certified Safety Professional (CSP) certified by the Board of Certified Safety Professionals or a licensed lead-based paint abatement Supervisor/Project Designer in the State of Maine.

1.2.12 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.13 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.14 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium (VI) or to plan such activities in commercial buildings, bridges or other structures.

1.2.15 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, and chromium (VI) to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.2.16 Engineering and Work Practice Controls

Controls that are structural or functional. Ventilation is an example of an engineering control. No eating or smoking in a LCC work area is a work practice control.

1.2.17 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, and chromium (VI)-contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.18 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. The use of the term Lead in this section also refers to paints which contain detectable concentrations of Cadmium and Chromium (VI). For the purposes of the section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium (VI).

1.2.19 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.20 Lead-Based Paint Activities

In the case of target housing or child occupied facilities, lead-based paint activities include; a lead-based paint inspection, a risk assessment, or abatement of lead-based paint hazards.

1.2.21 Lead-Based Paint Hazards

Paint-lead hazard, dust-lead hazard or soil-lead hazard as identified in 40 CFR 745, Section 65. Any condition that causes exposure to lead from lead-contaminated dust, lead-contaminated soil, lead-based paint that is deteriorated or present in accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects.

1.2.22 Lead, Cadmium, and Chromium (VI) Control Area

A system of control methods to prevent the spread of lead, cadmium, and chromium (VI) dust, paint chips debris, and/or any other associated materials to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.23 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8 hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 400/\text{No. hrs worked per day}$$

1.2.24 Material Containing Lead, Cadmium and Chromium (VI)/Paint with Lead, Cadmium, and Chromium (VI), (MCLCC/PWLCC)

Any material, including paint, which contains lead, cadmium or chromium (VI) as determined by the testing laboratory using a valid test method. The requirements of this section do not apply if no detectable concentrations of lead, cadmium, or chromium (VI) are found using an appropriate quantitative method for analyzing PCL or MCL using laboratory instruments with specified limits of detection. An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.25 Personal Sampling

Sampling of airborne lead, cadmium, and chromium (VI) concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 6 to 9 inches and centered at the nose or mouth of an employee.

1.2.26 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, and chromium (VI) control area to limit unauthorized entry of personnel.

1.2.27 Routes of Exposure

The way by which workers can come into contact, or be exposed, to a hazardous substance. Routes of exposure include inhalation (breathing in contaminated air, dust, vapors, or aerosols), ingestion (eating, drinking, chewing), injection (entering the body through a needle or sharp object puncturing the skin), absorption through the skin or eyes through direct contact.

1.2.28 Target Housing

Residential real property which is housing constructed prior to 1978, except housing for the elderly or persons with disabilities (unless any one or more children age 6 years or under resides or is expected to reside in such housing for the elderly or persons with disabilities) or any zero bedroom dwelling.

1.3 DESCRIPTION

This specification covers all project-related activities, including, but not limited to, renovation, construction, demolition, removal, and/or abatement, impacting PWL and/or material containing lead, cadmium, and chromium (VI). The work covered by this section includes work tasks and the precautions specified in this section for the protection of workers, building occupants, the building interior and exterior, furniture and equipment, and the environment during and after the performance of all project-related activities in accordance with state, local, and federal regulations and as specified herein.

1.3.1 Protection of Existing Areas To Remain

Any and all work performed under these specifications must be performed without damaging or impacting adjacent work and areas. Where such work or areas are damaged or impacted, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, and Chromium (VI) Compliance Plan and describe how the Contractor will prevent lead, cadmium and chromium (VI) exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium (VI) activities.

1.3.3 Sampling and Analysis

Submit all analytical results from sampling conducted during any and all work performed under these specifications, within the time-frames mandated by these specifications. Keep the log of results current with project activities and brief the results to the Contracting Officer as analytical results are reported.

Failure to submit the required data within the time frames detailed in the

specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and must not be made the subject, or any component of any request for additional time or compensation.

1.3.3.1 Dust Wipe Materials, Sampling and Analysis

Sampling must conform to ASTM E1728 and ASTM E1792. Analysis must conform to ASTM E1613 and ASTM E1644.

1.3.3.2 Soil Sampling and Analysis

Sampling must conform to ASTM E1727. Analysis must conform to ASTM E1613 and ASTM E1726.

1.3.3.3 Clearance Monitoring

Clearance sampling via air, wipe, and soil sampling must be performed in accordance with the requirements of Section entitled FIELD QUALITY CONTROL.

1.3.4 Clearance Requirements

Clearance requirements for air, wipe, and soil sampling are included in Section entitled in Section entitled FIELD QUALITY CONTROL.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Statement of Competent Person Assignment, and Competent Person Qualifications; G

Statement of CIH Assignment, and CIH Qualifications; G

Statement Identifying and Designating Roles and Responsibilities for CP, CIH, SSHO, and all Other Parties Involved in Heavy Metals Management; G

Training Certification; G

Occupational and Environmental Assessment Data Report; G

Medical Examinations; G

Lead, Cadmium, and Chromium (VI) Waste Management Plan; G

Licenses, Permits and Notifications; G

Occupant Protection Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist; G

Waste Characterization Sampling Plan; G

Analytical Results; G

SD-03 Product Data

Respirators; G

Vacuum Filters; G

Negative Air Pressure System; G

Materials and Equipment; G

Expendable Supplies; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Pressure Differential Log; G

SD-06 Test Reports

Sampling and Analysis; G

Occupational and Environmental Assessment Data Report; G

Sampling Results; G

Pressure Differential Recordings For Local Exhaust System; G

SD-07 Certificates

Testing Laboratory; G

Occupant Notification; G

Notification of the Commencement of Work Impacting Lead, Cadmium,
or Chromium (VI); G

Clearance Certification; G

SD-11 Closeout Submittals

Turn-In Documents or Weight Tickets; G

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Submit name, address, and telephone number of the CP selected to perform responsibilities specified in Section entitled COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related

experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, and chromium (VI); experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 5 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and licensed and certified in accordance with federal, State and local laws and regulations. The competent person must possess the qualifications detailed in Section entitled COMPETENT PERSON herein.

1.5.1.2 Training Certification

Submit a current certificate for each worker and supervisor, signed and dated by the accredited training provider, stating that the employee has received the required lead, cadmium and chromium (VI) training specified in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 745 and is certified to perform or supervise any and all work performed under these specifications.

1.5.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the air, soil, and wipe analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium (VI). Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved. Maine Department of Environmental Protection certification (ME L&C PROGRAM), National Environmental Laboratories Accreditation Conference (NELAC) certification, and certifications for all corresponding EPA test methods are required for waste determination testing, as applicable.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

- a. Verify training meets all federal, State, and local requirements.
- b. Review and approve Lead, Cadmium, and Chromium (VI) Compliance Plan for conformance to the applicable referenced standards.
- c. Continuously inspect LBP/PWL or MCL work for conformance with the applicable specifications and approved plan. Direct Contractor to immediately correct any non-conformance; Direct Contractor to Stop Work if non-conformance continues. Notify Contractor and Contracting Officer if non-conforming work is not corrected and/or stopped.
- d. Perform air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.
- e. Ensure work is performed in strict accordance with specifications at

all times. Direct Contractor to immediately correct any non conformance; Direct Contractor to Stop Work if non-conformance continues. Notify Contractor and Contracting Officer if non-conforming work is not corrected and/or stopped.

- f. Control work to prevent hazardous exposure to human beings, to adjacent areas, and to the environment at all times.
- g. Supervise final cleaning of the lead, cadmium, and chromium (VI) control area, take clearance wipe samples; review clearance sample results with respect to established clearance thresholds, determine requirements for additional cleaning, and advise the Contractor and Contracting Officer of the results and need for cleaning within the time-frames established in these specifications. Direct the Contractor to Stop Work in subject area pending achievement of clearance thresholds.
- h. Certify the conditions of the work as called for elsewhere in this specification.
- i. The CP must be certified pursuant to 40 CFR 745, Section 226 and is responsible for development and implementation of the occupant protection plan, all summary and status reports, and supervision of any and all work performed under these specifications.

1.5.2.2 Lead, Cadmium, and Chromium (VI) Compliance Plan

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium (VI), LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, and chromium (VI) control areas, critical barriers, physical boundaries, location and details of decontamination facilities, viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity of any and all work performed under these specifications. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, and chromium (VI) related work, collected waste water and dust containing lead, cadmium, and chromium (VI) and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, and chromium (VI) is not released outside of the lead, cadmium, and chromium (VI) control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksite's to inform affected employees and to clarify responsibilities to control exposures. If work will occur on a historic building, then include in the plan a description of how the proposed activities will comply with other relevant specifications to avoid or minimize impacts to historic fabric (for example, but not limited to, specification Sections related to historic masonry).

The plan must be developed and signed by a CIH, CSP, or, as appropriate, a certified Lead Project Designer in the State of Maine. The plan must include the name and certification number of the person signing the plan.

In occupied buildings, the plan must also include an occupant protection program that describes the measures that will be taken during the work to notify and protect the building occupants.

1.5.2.3 Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist

The Contractor must complete the Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist included as Attachment A at the end of this specification. The Checklist must be signed by the properly-certified person preparing the Plan and must be submitted to the Government for review and approval with the Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.5.2.4 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational and environmental sampling results to the Contracting Officer and PD&C HAZMAT POC within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 the Contractor must provide documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, and Chromium (VI) Compliance Plan.

- a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to estimate worker exposures per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, and chromium (VI) for stated work. The initial monitoring must represent each task and/or type of activity. The initial monitoring must represent each/every PWL and/or material containing lead, cadmium, and chromium (VI).
- b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 with a complete process description. This includes, but is not limited to, manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium (VI) containing coatings are present.
- c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, and chromium (VI) compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and

must not be made the subject, or any component of any request for additional time or compensation.

1.5.2.5 Medical Examinations

Submit a signed letter from an officer of the company confirming each worker engaged in heavy metals related activities in the contract has current medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Where exposure is determined to be above a lead, cadmium, or chromium (VI) action level the contractor must provide pre-work and post-work blood levels as requested by the Government. All medical records submitted must not contain any Protected Health Information (PHI) (e.g. home address, social security number, physicians name, medical record number, etc.).

Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, and chromium (VI) at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium (VI) in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, and chromium (VI) as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.6 Training

Train each employee performing any and all work performed under these specifications prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, 40 CFR 745, 49 CFR 172, and State of Maine and local regulations where appropriate.

1.5.2.7 Respiratory Protection Program

- a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and 29 CFR 1926.55.
- c. Provide training and fit test certificates for each employee performing LBP/ MCL/PWL work. Records submitted must not contain any Protected Health Information (PHI) (e.g. home address, social security number, physicians name, medical record number, etc.).

1.5.2.8 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.9 Lead, Cadmium, and Chromium (VI) Waste Management

The Lead, Cadmium, and Chromium (VI) Waste Management Plan must comply with applicable requirements of federal, State, and local hazardous and non-hazardous waste regulations and address:

- a. Identification and classification of wastes associated with the work.
- b. Estimated quantities of wastes to be generated and disposed of.
- c. Names and qualifications (experience and training) of personnel who will be working on-site with waste materials.
- d. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.
- e. Spill prevention, containment, and cleanup contingency measures including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.
- f. Work plan and schedule for waste characterization, containment and management. Proper containment of the waste includes using acceptable waste containers, as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.
- g. Include any process that may alter or treat waste. Note: altering or treatment of a hazardous waste is not allowed, including any process that dilutes the waste or renders a hazardous waste non-hazardous waste.
- h. All hazardous waste must be placed in a Shipyard-permitted Hazardous Waste Accumulation Area or Contractors must turn hazardous waste in to a Hazardous Waste Accumulation Area not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this contract will be considered Government-generated, and disposed of by the Government. Contractor must not bring hazardous wastes onto Government property.

1.5.2.10 Waste Characterization Sampling Plan

A sampling plan must be developed that details practices to be followed for the collection, management, and analysis of waste characterization samples to determine if waste materials generated by the referenced project will be classified as hazardous or non-hazardous to ensure proper disposal. The plan must propose sampling that meets the requirements of all applicable local, state, and federal statutes and regulations, Navy and PNSY requirements, and project specifications. At a minimum, the sampling plan must include:

- a. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen days from receipt of the plan.
- b. The timing of waste characterization sampling and analysis. To the extent feasible, waste sampling and analysis must be completed prior to the accomplishment of any project work to properly characterize

waste and ensure hazardous waste is not commingled with non-hazardous waste.

- c. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.
- d. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable including, but not limited to, 5090.6F and 5090.30.
- e. The proposed date(s) for sampling and expected duration of the sampling event(s). All waste sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.
- f. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
- g. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers. At a minimum, the waste characterization must include laboratory analysis for RCRA 8 Metals via the Toxicity Characteristic Leaching Procedure (TCLP). Depending on the methods used in project implementation, additional waste characterization analysis may be required (for example, waste characterization analysis for corrosivity may be required if high pH solutions are used for paint removal).
- h. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods.
- i. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.
 - 1. Laboratory analytical results.
 - 2. Field log book entries.
 - 3. Copy of the Chain of Custody (COC).
 - 4. Pertinent diagrams and maps.
 - 5. All other supporting data.

- j. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
- k. Confirmation that the sampling procedure will include, but not be limited to, the following:
 - 1. The Contractor will notify the Contracting Officer's Representative and PNSY Code 106.3 Sampling Program Manager seven (7) calendar days prior to each sampling event to ensure the area and materials to be sampled are witnessed by all parties.
- l. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
 - 1. The EFT taking the samples will inspect the work area/waste with Code 106.3 to visually and tactilely assess the materials to be sampled.
 - 2. Using a field log, the EFT will record the locations, types, quantities and other salient details related to materials to be sampled.
 - 3. The EFT will collect sufficient sample size of the materials from the project work area to complete all required analytical testing. Samples must be representative of all materials that comprise the waste for disposal.
 - 4. The EFT will place the sample in a clean jar or vial, secure the jar or vial closed and label it with a unique sample number.
 - 5. The EFT will record the sample number on the COC and note the location of the sample in the field log and on the site drawing.
 - 6. The EFT will verify that the sample number(s) on the jar/vial are accurate and consistent with the COC.
 - 7. The EFT will place the secured and labeled jar/vial into a cooler and arrange for prompt delivery of bulk samples to the analytical laboratory.
 - 8. The EFT will clean up any debris generated by the sampling process.
 - 9. The EFT will decontaminate the collection tool using wet wipes, water and paper towels after each sample is collected, or other approved method. Razor blades will be replaced on a periodic basis as determined by the EFT.
 - 10. All samples will be collected in a manner to eliminate cross-contamination by the previous sample.
 - 11. All materials used in the collection of the samples (gloves,

wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.

12. The EFT will obtain the following sampling equipment to perform the bulk sampling.

- (a) Individual sample jars and vials;
- (b) Field log;
- (c) Site diagram;
- (d) Chain-of-Custody (COC) forms;
- (e) Nitrile gloves;
- (f) Stainless steel or plastic soil scoops;
- (g) Metal chisels of assorted sizes;
- (h) Razor knives and replacement blades;
- (i) Water and spray bottle;
- (j) Trash bags;
- (k) Disposable cleaning wipes;
- (l) Paper towels;
- (m) Duct tape;
- (n) Flashlight;
- (o) Personnel lifts; and
- (p) Other equipment as necessary.

1.5.2.11 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of federal, State, and local authorities regarding lead, cadmium and chromium. Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, EM 385-1-1, and ND OPNAVINST 5100.23. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply.

1.5.3 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, and chromium (VI) control area is removed. Submit pressure differential recordings for each work day to the Contracting Officer and PD&C HAZMAT POC within 24-hours from the end of each work day.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and must not be made the subject, or any component of any request for additional time or compensation.

1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Maine Department of Environmental Protection and the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits, and notifications have been

obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the Contract price.

1.5.5 Occupant Protection Plan

The CP must develop and implement an Occupant Protection Plan describing the measures and management procedures to be taken during any and all work performed under these specifications to protect the building occupants/building facilities and the outside environment from exposure to any lead, cadmium and chromium (VI) impacts while any and all work is performed under these specifications.

1.5.6 Hazardous Materials Meeting

Conduct a safety preconstruction conference to discuss the details of the Lead, Cadmium, and Chromium (VI) Waste Management Plan including procedures and precautions for the work, roles and responsibilities, monitoring and sampling requirements, and Stop Work conditions. Also, to discuss the Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). The safety preconstruction conference must include NAVFAC Lead Program Manager, the Contractor and their Designated Competent Person, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium (VI)dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, and chromium (VI) must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during any and all work performed under these specifications, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

Utilize UL-Labeled HEPA filters for any and all work performed under these specifications.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal

protective equipment (PPE) daily, as required herein, for entry into and inspection of any and all performed under these specifications work within the lead, cadmium and chromium (VI) controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machines for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers and the Contracting Officer. Documentation of CR Manager approval must be included in the 'Lead, Cadmium, and Chromium (VI) Compliance Plan.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of ASSP/SAFE Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of negative 0.02 inch of water column relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

- a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.
- b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.
- c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 2.5 inches of water static pressure differential on a magnehelic gauge.
- d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.02 inch of water and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.
- e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.
- f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.
- g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.

- h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 1,100 degrees F. Heat blower must be (grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that contamination may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection including all necessary wipe and air sampling results are submitted to the Contracting Officer directly from the CP, work may proceed at the discretion of the Contracting Officer. All cleanup and testing must be completed in accordance with the requirements outlined herein this specification. All costs for the required clearance sampling and clean up efforts are the responsibility of the Contractor. Any delays associated with the clean up efforts are the responsibility of the Contractor. Achievement of restoration must be documented via air, wipe and/or soil sampling, as required under the Section entitled FIELD QUALITY CONTROL.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 6 mils thick (4 mils thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, and chromium (VI) contaminated debris must be polyethylene plastic that is a minimum of 6 mils thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 6 mil, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out-of-the-ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 2 inches wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT and OSHA standards. For containerization of hazardous waste, the Contractor must use containers provided by B357.

2.1.1.6 Chemical Paint Strippers

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Neutralizers for paint strippers must be compatible with the substrate and

suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium (VI) control work using cleaning techniques specified by HUD 6780 guidelines.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Notification

- a. Notify the Contracting Officer 20 calendar days prior to the start of any and all work performed under these specifications.
- b. Occupant Notification

Provide building occupants with copy of Occupant Protection Plan prepared under Section entitled OCCUPANT PROTECTION PLAN. Where appropriate, submit occupant written acknowledgment of the delivery of lead hazard information pamphlet (EPA 747-K-99-001 "Protect Your Family From Lead in Your Home") prior to commencing the renovation work for each affected unit using language provided in 40 CFR 745 Subpart E.

3.1.1.2 Lead, Cadmium, and Chromium (VI) Control Area

- a. Physical Boundary - Provide physical boundaries around the lead, cadmium, and chromium (VI) control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium (VI) will not escape outside of the lead, cadmium and chromium (VI) control area. Prohibit all personnel, except those specifically authorized to enter, from accessing the lead, cadmium, and chromium (VI) control areas.
- b. Warning Signs - Provide warning signs at approaches to lead, cadmium, and chromium (VI) control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.3 Furniture and Equipment

Furniture and equipment will remain in the lead, cadmium, and chromium (VI) control area. Protect and cover furnishings and equipment or remove furnishings and equipment from the work area and store in a location approved by the Contracting Officer.

3.1.1.4 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, and chromium (VI) control areas. Seal intake and exhaust vents in the lead, cadmium, and chromium (VI) control area with 6 mil plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, and chromium (VI) control area. Provide temporary HVAC system for areas in which HVAC has been shut down outside the lead, cadmium, and chromium (VI) control area.

3.1.1.5 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ASSP/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until all wipe testing clearance thresholds have been achieved and the lead, cadmium, and chromium (VI) control area is removed. The enclosure must be leak proof and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of negative 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ASSP/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.6 Negative Air Pressure System Containment

- a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until all wipe testing clearance thresholds have been achieved and the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor, under the oversight of the CP, is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Stop work and notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium (VI) contaminated waste.
- b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during work activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Protocols for dismantling of the negative air pressure systems must be as presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination. Failure to submit the required data within the time frame detailed in the specifications will be considered non-compliance.

with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and must not be made the subject, or any component of any request for additional time or compensation.

3.1.1.7 Decontamination Shower Facility

If the potential for exceedance of the PEL for lead, cadmium and/or chromium (VI) exists during work performed under these specifications, the decontamination facility must include a shower.

Provide clean and contaminated change rooms and shower facilities in accordance with this specification and 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

3.1.1.8 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.9 Mechanical Ventilation System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ASSP/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until all wipe testing clearance thresholds have been achieved and the lead, cadmium, and chromium (VI) control area is removed. The enclosure must be leak proof and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of negative 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ASSP/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.10 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, and chromium (VI) control area. No one will be permitted in the lead, cadmium, and chromium (VI) control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, and Chromium (VI) Control Area Requirements

Establish a lead, cadmium, and chromium (VI) control area by completely establishing barriers and physical boundaries as appropriate around the area or structure where PWLCC or MCLCC activities will be performed.

Full containment - Contain operations by the use of critical barriers and HEPA filtered exhaust, or a negative pressure enclosure system with decontamination facilities and with HEPA filtered exhaust where work practices will create airborne dust containing lead, cadmium, chromium (VI) (such as sanding, sawing, grinding, thermal cutting, or digging or demolition activities) as required by the CP. For containment areas larger than 1,000 square feet install a minimum of two 18-inch square viewing ports. Locate ports to provide a view of the required work from the exterior of the enclosed contaminated area. Glaze ports with laminated safety glass.

3.3 APPLICATION

3.3.1 Lead, Cadmium, and Chromium (VI) Work

Perform any and all work under these specifications in accordance with approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Use procedures and equipment required to limit occupational exposure, impacts to building interiors/exterior, and environmental contamination with lead, cadmium, and chromium (VI) when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 or 40 CFR 745 and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with these specifications, Navy and PNSY requirements, and federal, State, and local requirements.

3.3.2 Removal of Materials with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI)

Manual or power sanding or grinding of lead, cadmium, and chromium (VI) surfaces or materials is not permitted unless using wet methods and/or tools equipped with HEPA attachments. The dry sanding or grinding of surfaces that contain lead, cadmium, and chromium (VI) is prohibited. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers and the Contracting Officer. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Provide methodology for removing lead, cadmium, and chromium (VI) in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Select lead, cadmium, and chromium (VI) removal processes to prevent contamination of work areas outside the control area and/or areas that are required to be protected (i.e. HVAC systems, furniture, equipment, etc.), with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris or waste and to ensure that unprotected personnel are not exposed to concentrations of lead, cadmium, and chromium (VI) exceeding the applicable Action Levels. Describe this removal process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

While performing this work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3.2.1 LCC Work Categories

The purpose of this section is to provide work control practices for workers to minimize occupational exposure to LCC while working with Material with Lead, Cadmium and Chromium (VI) and to ensure that these practices are consistent with OSHA standards (29CFR§1926.62, §1926.1126, and §1926.1127). This applies to LCC operations conducted at Portsmouth Naval Shipyard (PNS) and other Areas of Responsibility (AOR) sites.

These categories are being established to define the general level of controls, monitoring, training, and other requirements associated with different work activities commonly performed on projects at PNSY and other AOR sites. These categorizations are intended to be general guidelines only and do not absolve the contractor from responsibility for performing an assessment of the materials, conditions, and activities and determining the appropriate worker safety requirements.

3.3.2.1.1 Category A LCC Work

This category is reserved for LCC operations expected to result in exposure below the Action Limit. Category A operations will require an initial industrial hygiene assessment and whenever the operation changes. Category A work typically requires awareness level training but does not require monitoring, use of engineering controls, or respiratory protection. The work must be documented in an Activity Hazard Analysis form and a wash station must be provided for workers to wash hands and face after completion of work. Dry sweeping and use of compressed air for cleanup is prohibited. After leaving the work area, workers must wash their hands and faces prior to eating, drinking, smoking or applying cosmetics. Eating, drinking, chewing, smoking, etc. is not permitted within the lead work area.

Typical operations that could be considered Category A tasks include work with materials where no heavy metals are detected, limited drilling or sawcutting activities of concrete coated with LCC paint where controls are already being employed to comply with the OSHA silica standard, whole component removal not subject to sawing, grinding, drilling or other aggressive methods, and application of chemical stripper in accordance with manufacturer's recommendations.

3.3.2.1.2 Category B LCC Operations

This category is reserved for LCC operations expected to result in exposure at or above the Action Limit but below the Permissible Exposure Limit. Category B operations will require an initial industrial hygiene assessment and whenever the operation changes. Category B work typically requires adequately trained personnel, periodic IH monitoring (initial and weekly thereafter), isolation/demarcation of regulated areas, may require use of wet methods and HEPA equipped vacuums, personal protective equipment, and suitable decontamination facilities. Dry sweeping and use of compressed air for cleanup is prohibited. After leaving the work area, workers must wash their hands and faces prior to eating, drinking, smoking or applying cosmetics. Eating, drinking, chewing, smoking, etc. is not permitted within the lead work area. This work must be documented in a LCC compliance work plan in accordance with the appropriate section of this specification.

Typical operations that could be considered Category B tasks include drilling or sawcutting surfaces where LCC paint or dust is present without

the use of wetting or HEPA equipped vacuum shrouds, removal of chemical strippers with LCC paint, surface cleaning where LCC materials are present, and exterior paint scraping operations.

3.3.2.1.3 Category C LCC Operations

This category is reserved for LCC operations expected to result in exposure above the permissible exposure limit (PEL). Category C operations will require an initial industrial hygiene assessment and whenever the operation changes. Category C work requires adequately trained personnel, daily IH monitoring, isolation/demarcation of regulated areas, negative pressure enclosures, use of wet methods and HEPA equipped vacuums, personal protective equipment, and 3-stage decontamination facilities with shower. Dry sweeping and use of compressed air for cleanup is prohibited. This work must be documented in a LCC compliance work plan in accordance with the appropriate section of this specification.

Typical operations that could be considered Category C tasks include interior scraping of LCC paint, dry hand scraping or sanding of LCC paint, abrasive blasting, grinding, burning or other hot work on LCC paint and other LCC materials.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, and chromium (VI) controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

- a. Vacuum all clothing before entering the contaminated change room.
- b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.
- c. Shower.
- d. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air, Wipe and Soil Sampling Before and During Work

Conduct sampling for lead, cadmium, and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 and as specified herein. Air, wipe and soil sampling must be directed or performed by the CP.

- a. The CP must be on the job site directing and/or performing the air, soil and wipe sampling and inspecting the project work to ensure that the requirements of the contract, local, state and federal statutes and regulations, Navy and PNSY requirements, and project specifications have been satisfied during all work performed under these specifications.
- b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew

or a minimum of two employees, whichever is greater, for each task or activity with a minimum of two rounds of testing.

- c. Conduct area air sampling as depicted above in areas adjacent to the lead, cadmium and chromium (VI) control area, at exhaust points for negative air machines, at entrances to the negative pressure enclosure (NPE), and at other locations determined appropriate for the project and site conditions. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above the applicable Action Levels. If the Action Level is reached or exceeded, stop work, correct the conditions(s) causing the increased levels if known, and notify the Contracting Officer immediately. Determine if condition(s) require any further change in work methods. Resume LCC work only after the CP and the Contracting Officer give approval.
- d. Submit the analytical results for the personal and area air monitoring samples, signed by the CP, to the Contracting Officer within 72-hours after the air samples are collected.
- e. Before any work begins, the CP must collect and analyze baseline wipe, air, and soil (if bare soil is present during exterior remediation operations) samples in accordance with methods defined by federal, State, and local standards inside and outside of the physical boundary to establish baseline conditions in the facility prior to performance of any and all work under these specifications. Provide Initial Sample Results to the Contracting Officer before work begins.
- f. Surface Wipe Samples- If at any time lead, cadmium and/or chromium (VI) concentrations in area air monitoring samples exceed Action Levels, stop work and collect surface wipe samples exterior to the work area. Samples must be collected at a sufficient frequency to properly evaluate area conditions. Surface wipe sample results must meet criteria in Section entitled CLEARANCE CERTIFICATION. If sample results exceed clearance criteria, then the area must be cleaned until the criteria is met. Work may recommence when the criteria are met with approval from the Contracting Officer.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and must not be made the subject, or any component of any request for additional time or compensation.

3.4.1.2 Wipe and Soil Sampling After Work

The clearance thresholds for required wipe sampling are as follows:

- (1) Lead- 40 ug/sq.ft. on floors; 250 ug/sq.ft. on interior window sills; 400 ug/sq. ft. on window troughs; or below concentrations detected in wipe samples collected before commencement of work - whichever is lower.
- (2) Chromium (VI)- Below concentrations detected in wipe samples collected before commencement of work.
- (3) Cadmium- Below concentrations detected in wipe samples collected before commencement of work.

For any and all work performed under these specifications, surface wipe sampling must be conducted and clearance determinations made according to the requirements detailed in Section entitled SAMPLING AND ANALYSIS herein and to the work practice standards presented in 40 CFR 745.227.

3.4.1.3 Testing of Material Containing Lead, Cadmium, and Chromium (VI) Residue

Test residue in accordance with 40 CFR 261 for hazardous waste.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, and chromium (VI) control area free of accumulations of dust and debris. Restrict the spread of dust and debris to other areas within the facility; keep waste from being distributed over & beyond the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the the work performed under these specifications has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area in accordance with these specifications and the approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become impacted at any time during the work, clean, visually inspect, and then wipe sample all impacted areas. The CP must then certify in writing to the Contracting Officer that the area has been cleaned of lead, cadmium and chromium (VI) impacts before clearance testing.

3.5.1.1 Clearance Certification

The CP must certify in writing to the Contracting Officer and the PD&C HAZMAT POC that air samples collected outside the lead, cadmium, and chromium (VI) control area, including the Facility's HVAC Ductwork and Filters, during performance of any and all work under these specifications are less than the applicable Action Levels of 30 micrograms of lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; that there were no visible accumulations of material and dust containing lead, cadmium, and chromium (VI) left in the work site, and all applicable wipe and soil clearance sample analytical results are below applicable clearance thresholds. The Contractor is to clean all HVAC Ductwork if the facility's HVAC system is operational and cannot be isolated from the work control area or there is any evidence that the pre-filters were not effective. Do not remove the lead, cadmium, and chromium (VI) control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt and approval of the CP certification. Clearance sampling results must be submitted to the Contracting Officer within 72 hours after the samples are taken. The Contractor should allow fourteen (14) calendar days for the Government review period.

3.5.2 Disposal

- a. Dispose of material, whether hazardous or non-hazardous in accordance with local, state, and federal Statutes and Regulations, Navy and PNSY requirements, and project specifications. Ensure all waste is properly characterized in accordance with the Waste Characterization Sampling Plan developed under paragraph entitled WASTE CHARACTERIZATION SAMPLING PLAN herein. The result of each waste characterization will dictate disposal requirements.
- b. Contractor is responsible for segregation of waste. Collect lead, cadmium, and chromium (VI) contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, and chromium (VI)-contaminated clothing. Label the containers in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.
- c. Accumulate waste materials in U.S. Department of Transportation (49 CFR 178) approved 55 gallon drums or appropriately sized container for smaller volumes. Properly label each drum to identify the type of hazardous material (49 CFR 172). For hazardous waste, the collection container requires marking/labeling in accordance with 40 CFR 262 and corresponding state regulations.
- d. Manage lead, cadmium, and chromium (VI) or lead, cadmium, and chromium (VI) contaminated waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265.
- e. All lead, cadmium, and chromium (VI) waste generation, management, and disposal will be coordinated with the Shipyard environmental function (Code 106, B357) and the Contracting Officer.

All hazardous waste must be turned in to B357 not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this contract will be considered Government-generated, and disposed of by the Government. Contractor must not bring hazardous wastes onto Government property.

Containers positioned within the work area boundaries must have covers in place whenever containers are not in use.

3.5.2.1 Disposal Documentation

For non-hazardous waste, provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

3.5.2.2 Payment for Non-Hazardous Waste

Payment for disposal of non-hazardous waste will not be made until a signed copy of the manifest from the disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of non-hazardous waste delivered to the disposal facility.

-- End of Section --